Case Study 3: Fraud Detection in Online Banking

Dataset Source: *Kaggle – Credit Card Fraud Detection Dataset*

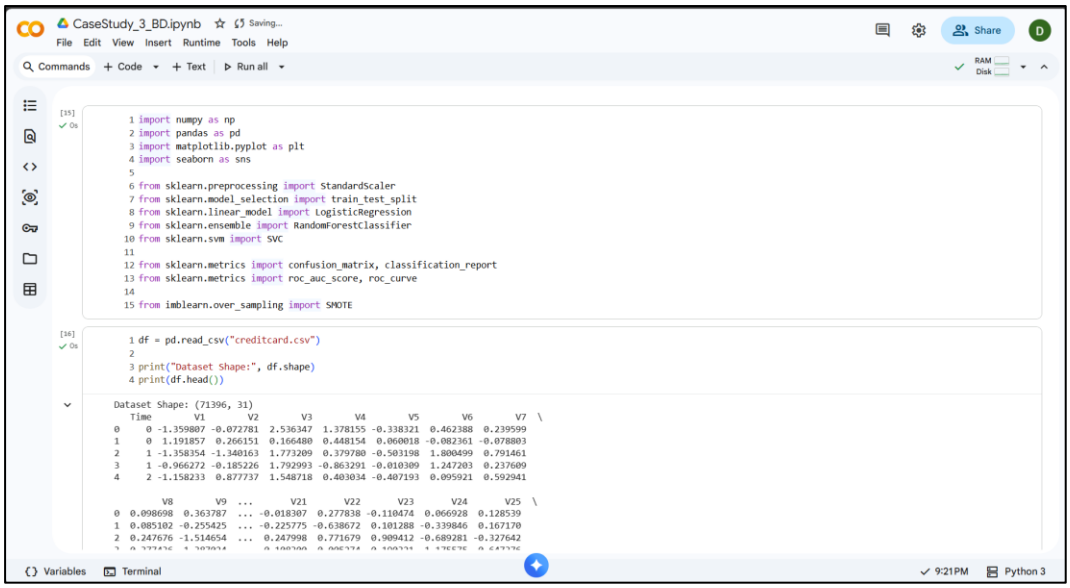
1. Background

A bank is experiencing an increase in fraudulent online transactions. Fraud transactions lead to financial losses, reduced customer trust, and legal risks.

The bank collects transaction data including:

- Transaction Amount
- Time of Transaction
- Device / Transaction behaviour
- Customer Spending History
- Merchant Type (conceptual in real banking systems)

Using this data, the goal is to detect fraudulent transactions using machine learning techniques.



2. Problem Statement

Fraud detection is defined as a **binary classification problem**:

- **Class 0 → Legitimate Transaction**
- **Class 1 → Fraudulent Transaction**

Goal:

- Detect maximum fraud cases
- Minimize false alarms
- Maintain high model accuracy

3. Class Imbalance Issue

In real banking systems, fraud transactions are extremely rare.

From the dataset:

- Total transactions = **71396**
- Fraud transactions = **31**
- Fraud Percentage ≈ **0.25%**
- Genuine transactions = **99.75%**

This creates a **high class imbalance problem**.

Why is this dangerous?

If 99% transactions are genuine:

A model predicting everything as genuine will show:

99% Accuracy
0% Fraud Detection

Therefore, **accuracy alone is misleading**.

4. Data Processing

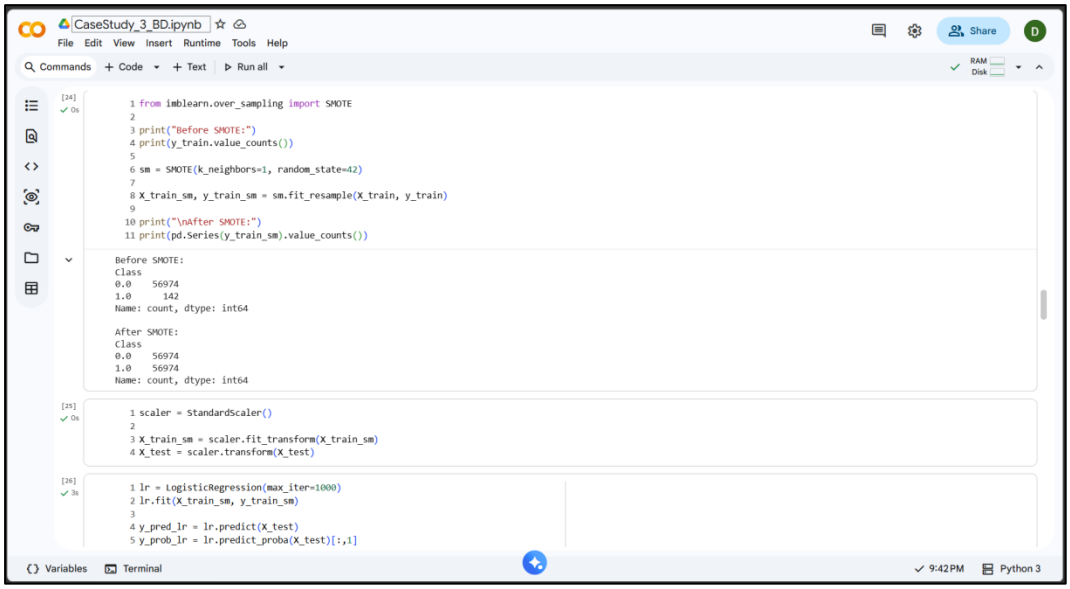
- **Handling Imbalanced Data (SMOTE)**

We use **SMOTE (Synthetic Minority Over-sampling Technique)**.

Concept:

- Generates synthetic fraud samples
- Balances minority class
- Applied only on training data

Why? To prevent model bias toward majority class.



5. Feature Engineering

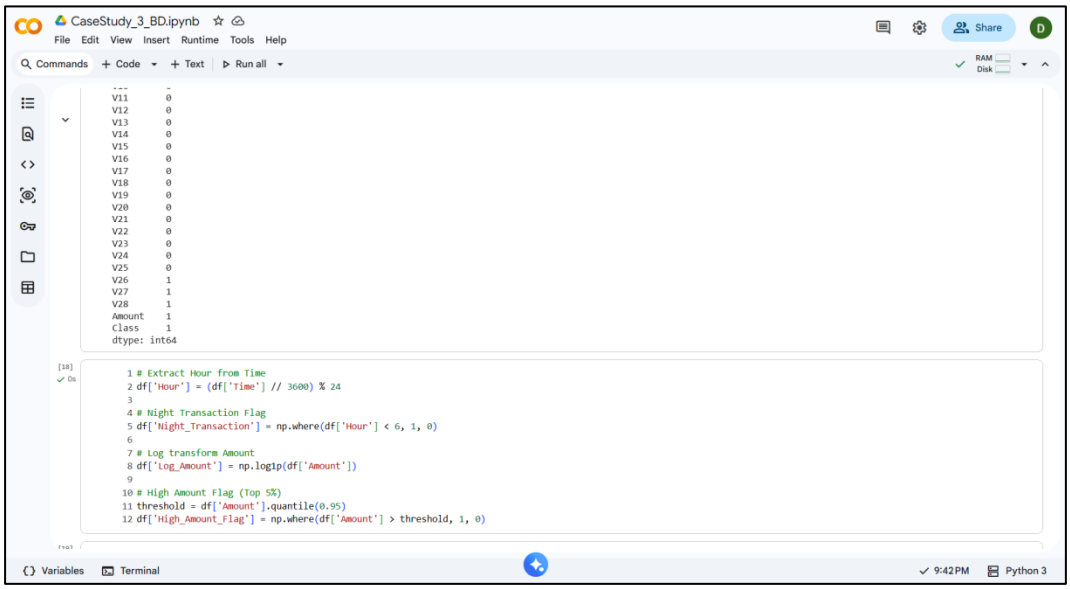
Feature engineering improves fraud detection performance.

Since Kaggle dataset contains:

Time ,Amount ,PCA features (V1–V28) ,Class

Engineered features:

1) **Transaction Hour -** Fraud often occurs at unusual hours
2) **Night Transaction Flag -** 12 AM–6 AM flagged as suspicious
3) **Log Amount -** Reduces skew in transaction amounts
4) **High Amount Flag -** Transactions above 95th percentile marked high risk.



6. Descriptive Analytics

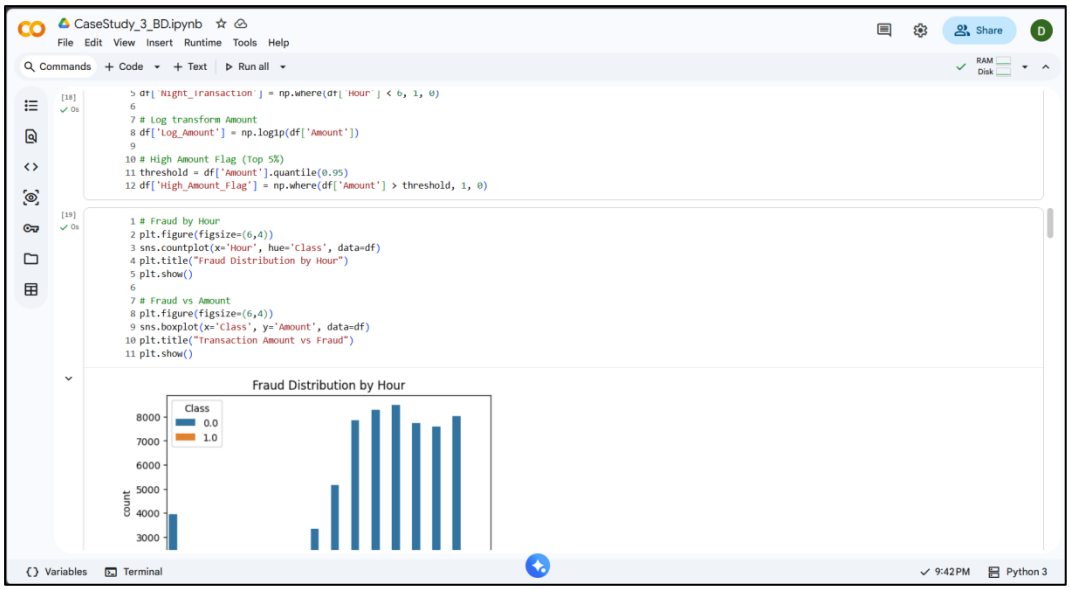
Before modeling, fraud patterns were analyzed.

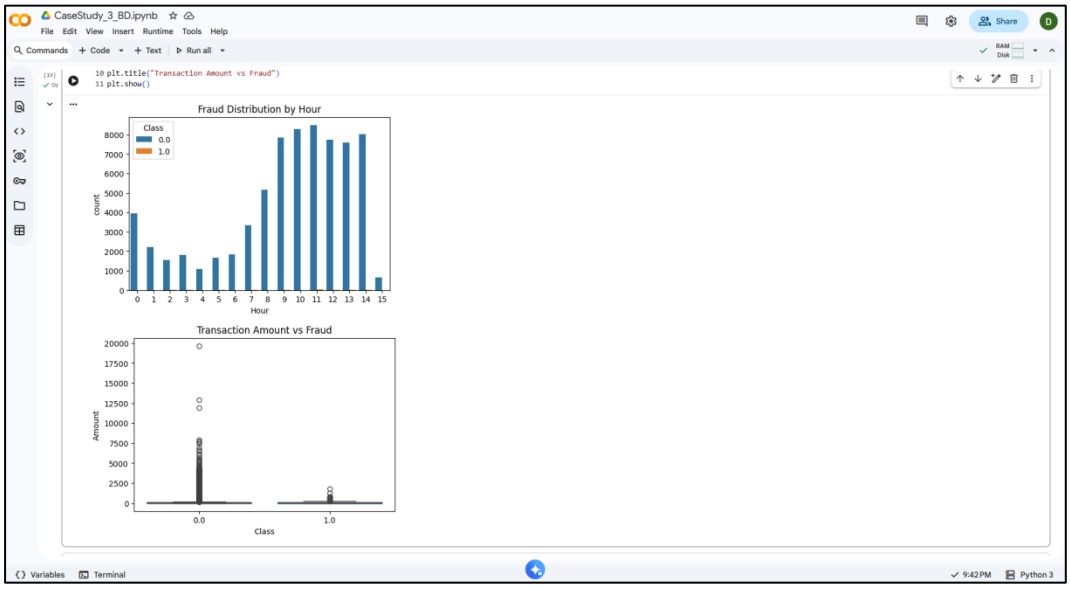
Fraud by Time of Day - Night transactions showed higher fraud probability.

Visualization: Bar Plot

Fraud by Transaction Amount - Fraud occurs in Very small test transactions or Very large suspicious transfers

Visualization: Boxplot





7. **Evaluation Metrics**
   - **Confusion Matrix**

     Shows: True Positive (TP) ,False Positive (FP) ,False Negative (FN) ,True Negative (TN)

   - **Precision**

Precision = TP / (TP + FP)

Out of predicted fraud, how many were actually fraud? Important to reduce false alarms.

- **Recall (Most Important)**

  Recall = TP / (TP + FN)

  Out of actual fraud, how many did we detect?

  In fraud detection: False Negative is more dangerous than False Positive.

  If fraud is predicted as genuine:

  - Bank loses money
  - Customer trust reduces
  - Legal risks increase

  Therefore: High Recall is prioritized.

- **F1 Score**

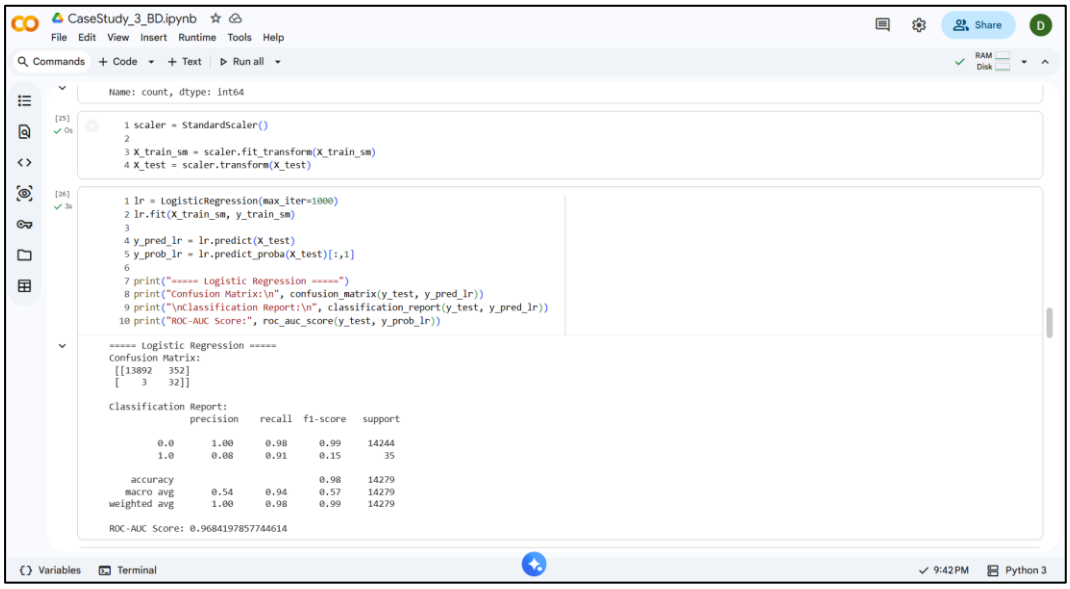
  Balances precision and recall.

8. Predictive Models Used

Three classification algorithms were applied:

1) Logistic Regression

- Baseline model
- Works well with linear separation
- Fast and interpretable



Logistic Regression Result Analysis:

- **Confusion Matrix**

	Predicted Genuine	Predicted Fraud
Actual Genuine	13892	352
Actual Fraud	3	32

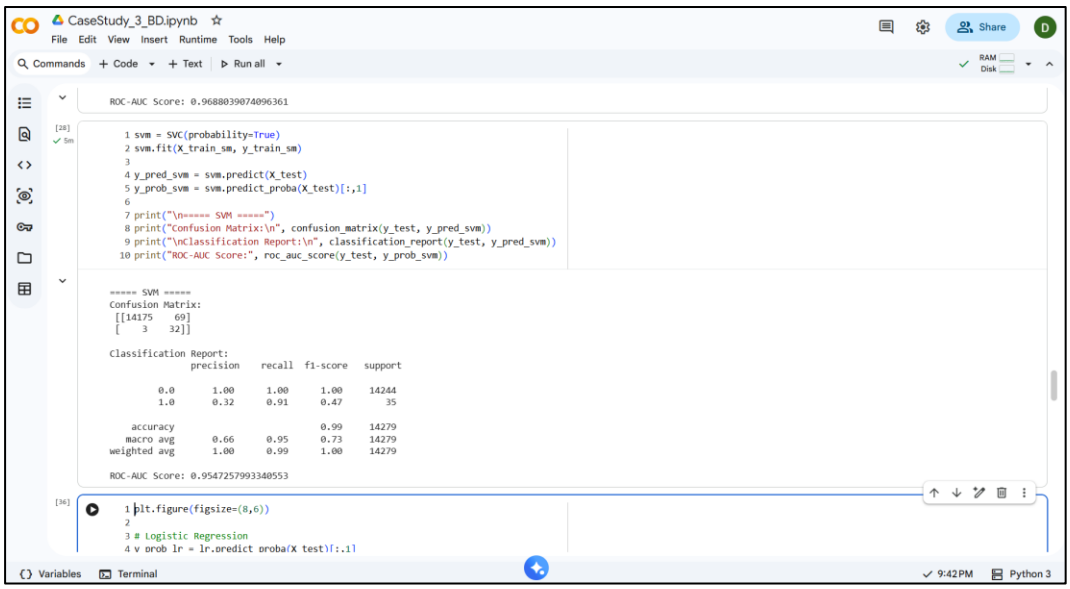
Metric	Value
Fraud Detected	32 / 35
Missed Fraud	3
Recall	0.91
Precision	0.08
Accuracy	0.98
ROC-AUC	0.968

Is This Model Good?

High recall – most fraud detected
Low precision – many false alarms due to rare fraud
Good overall for banking, prioritizes fraud detection

2) Support Vector Machine (SVM)

- Works well in high-dimensional space
- Finds optimal separating hyperplane
- Handles complex fraud boundaries



SVM Result Analysis:

- **Confusion Matrix**

	Predicted Genuine	Predicted Fraud
Actual Genuine	14175	69
Actual Fraud	3	32

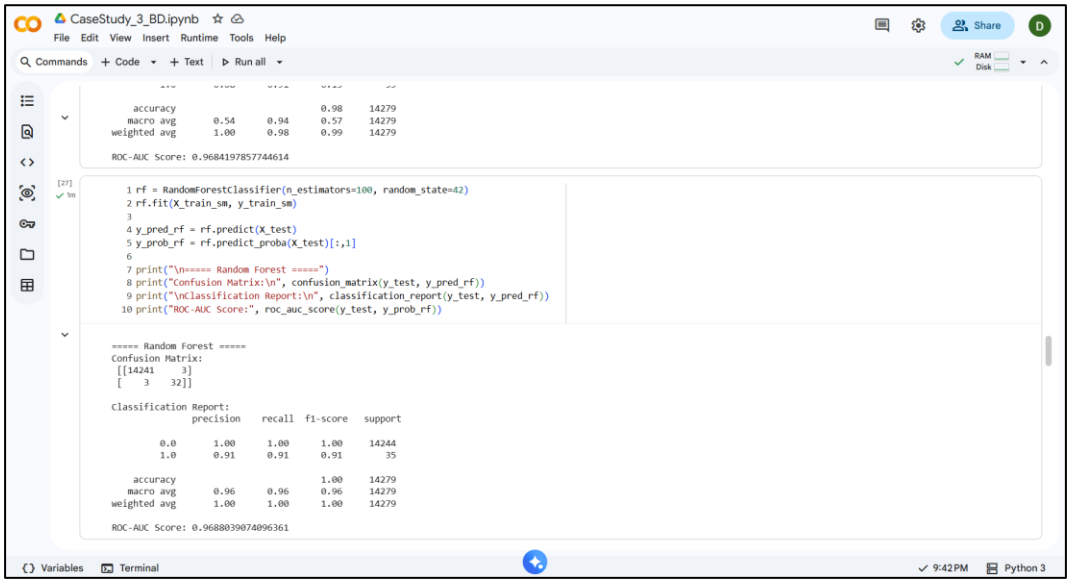
Metric	Value
Fraud Detected	32 / 35
Missed Fraud	3
Recall	0.91
Precision	0.32
Accuracy	0.99
ROC-AUC	0.955

Is This Model Good?

High recall, moderate precision
Some false alarms, less balanced than Random Forest

3) Random Forest (Best Performer)

- Ensemble of decision trees
- Handles non-linearity
- Reduces overfitting
- Automatically detects feature importance



Random Forest Result Analysis (Best Performer):

- **Confusion Matrix**

	Predicted Genuine	Predicted Fraud
Actual Genuine	14241	3
Actual Fraud	3	32

Metric	Value
Fraud Detected	32 / 35
Missed Fraud	3
Recall	0.91
Precision	0.91
Accuracy	1.0
ROC-AUC	0.9688

Is This Model Good?

High recall & precision
Minimal false alarms
Best balance, ideal for fraud detection

Model	Precision	Recall	F1-Score	Accuracy	ROC-AUC	Comments
Logistic Regression	0.08	0.91	0.15	0.98	0.968	High recall, low precision
SVM	0.32	0.91	0.47	0.99	0.955	Moderate precision, more false alarms
Random Forest	0.91	0.91	0.91	1.00	0.969	Best balance, ideal for fraud detection

9. Why Ensemble Models Perform Better

Fraud patterns are:

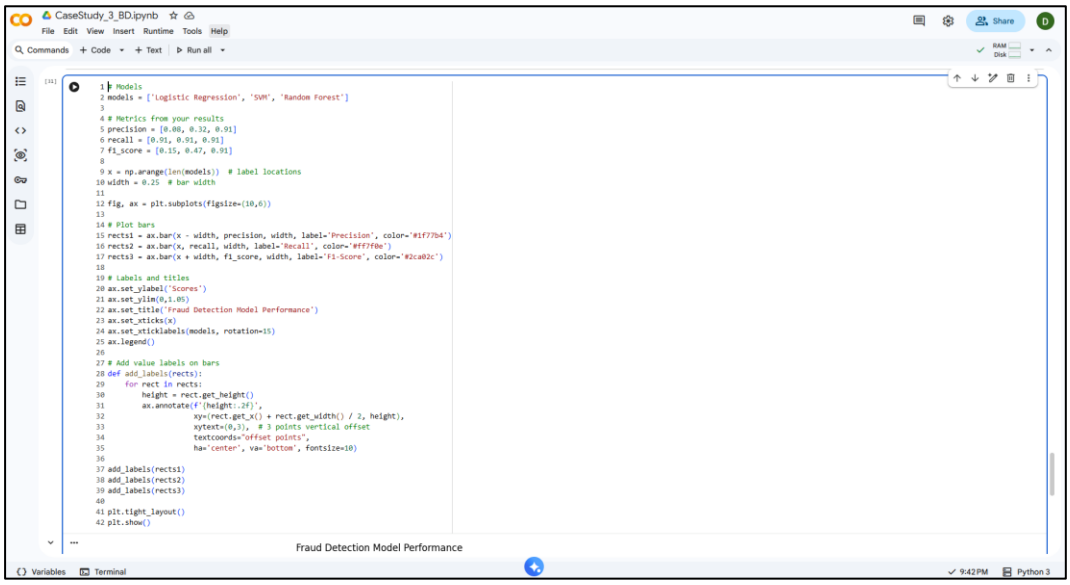
Complex, Dynamic, Non-linear, Rare

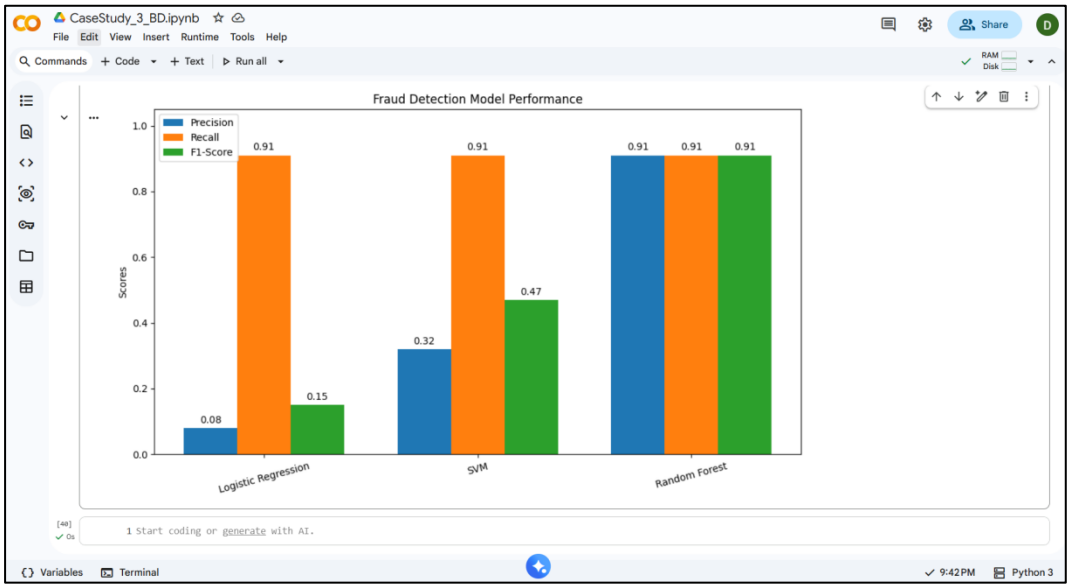
Ensemble models like Random Forest:

- Combine multiple trees
- Reduce variance
- Capture complex patterns

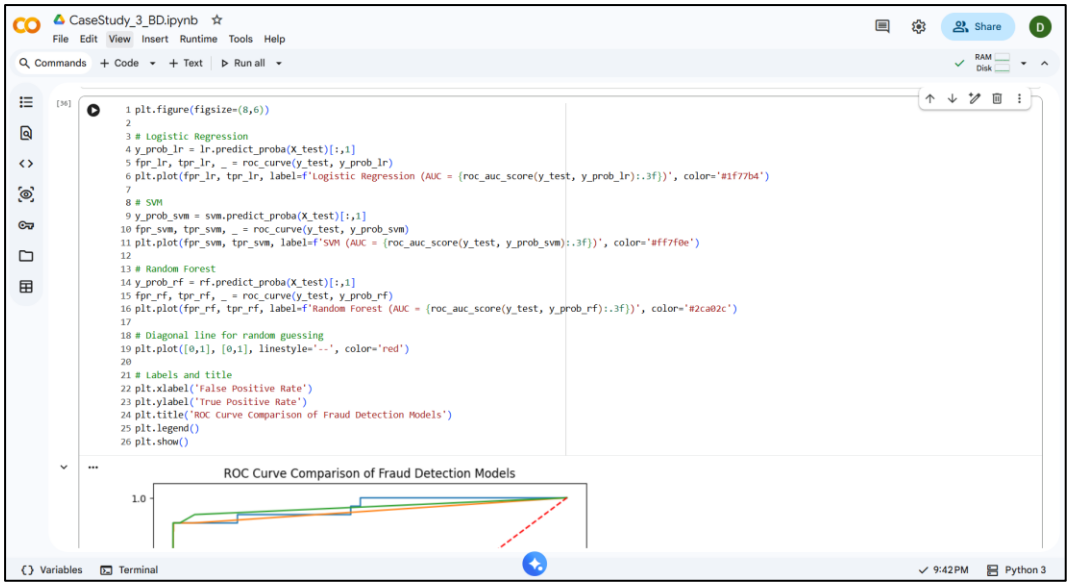
- Generalize better
- More robust to noise

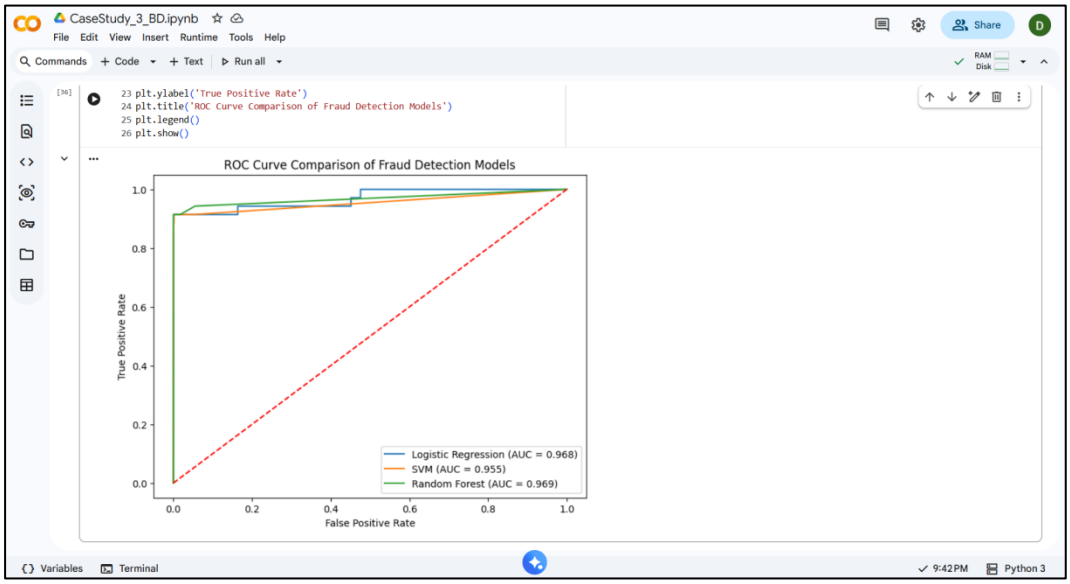
Therefore, Random Forest usually performs best.





- **ROC Curve & AUC**
  - Plots True Positive Rate vs False Positive Rate
  - AUC close to 1 → strong model





10. Question:

Why is recall more important than accuracy in fraud detection?

- Fraud is **rare** (~0.25%), so high accuracy can be misleading.
- **Recall** tells us: "Out of all real frauds, how many did we catch?"
- **Missing a fraud (False Negative)** = Bank loses money, customer trust drops, legal risks rise.
- **False alarms (False Positive)** = Minor inconvenience, usually acceptable.
- In fraud detection, **catching fraud matters more than overall accuracy**.